Project Report: TF-IDF Based Document Search Engine

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Course: CSA2001 - Fundamentals in AI & ML

1. The Problem, Its Importance, and Relevance to AI & ML

With the exponential growth of digital text data, finding relevant information quickly and accurately is a significant challenge. Simple keyword matching is often insufficient because it doesn't account for the importance or rarity of words across a large corpus of documents. The problem addressed in this project is **Information Retrieval**: how to efficiently search through a large dataset of text documents and return the most relevant results based on a user's query.

Relevance to AI & ML: This project directly applies core concepts taught in the *Fundamentals of AI & ML* coursework, specifically within the domain of Natural Language Processing (NLP) and Unsupervised Learning techniques. Machine learning algorithms cannot process raw text; they require numerical input. By implementing a vector space model, this project demonstrates **feature extraction and engineering**—a critical first step in any ML pipeline. Transforming unstructured text into structured, mathematical representations (sparse matrices) bridges the gap between human language and computational logic. Furthermore, utilizing **Cosine Similarity** introduces foundational concepts of distance metrics and vector mathematics, which are the underlying mechanics for many AI applications, including recommendation systems, clustering (like K-Means), and advanced neural networks.

2. Approach to Solving the Problem

To solve this, I implemented a vector space model using the **TF-IDF (Term Frequency-Inverse Document Frequency)** statistical measure.

- **Dataset:** I used the standard `20 Newsgroups` dataset, a collection of approximately 20,000 newsgroup documents partitioned across 20 different topics.
- **Text Vectorization (`IDF-matrix_updated.py`):** Raw text cannot be processed directly by mathematical models. I utilized a TF-IDF Vectorizer to convert the text documents into a sparse matrix. This technique evaluates how relevant a word is to a document in a collection. It increases proportionally to the number of times a word appears in the document but is offset by the number of documents that contain the word.
- **Search Mechanism (`IDF-search_updated.py`):** When a user enters a query, the system transforms the query using the *same* TF-IDF vectorizer vocabulary. It then

calculates the **Cosine Similarity** between the query vector and all document vectors in the dataset to find the closest matches.
- **Model Persistence:** To make the application efficient, I serialized (saved) the computed TF-IDF matrix, the vectorizer, and target mappings using `joblib`. This ensures that the heavy computation is done only once, and subsequent searches load the pre-computed models instantly.

3. Key Decisions

- **Choosing TF-IDF over Bag of Words (BoW):** While BoW counts word frequencies, it gives undue weight to common words (like "the", "is", "and"). I chose TF-IDF because it penalizes highly frequent words across the corpus, allowing unique, distinguishing words to drive the search results.
- **Using `joblib` for Serialization:** I opted to save the model states as `.joblib` files rather than `.pkl` (pickle) files. `Joblib` is specifically optimized for saving large NumPy arrays and SciPy sparse matrices, which perfectly suits the large TF-IDF sparse matrix generated from the 20 Newsgroups dataset.

4. Challenges Faced

- **Vocabulary Alignment:** A major challenge was ensuring that the search query was transformed using the exact same feature space (vocabulary) as the original dataset. If the vectorizer wasn't saved and loaded correctly, the matrix dimensions wouldn't match, causing the cosine similarity calculation to fail. This was resolved by meticulously exporting the fitted `tfidf_vectorizer_verified.joblib`.
- **Handling High Dimensionality:** The 20 Newsgroups dataset contains a massive vocabulary, resulting in a very wide sparse matrix. Managing memory and ensuring the cosine similarity calculations remained fast required using efficient SciPy sparse matrix operations rather than standard dense arrays.

5. What I Learned

Through this project, I gained hands-on experience with Natural Language Processing (NLP) pipelines. I learned how to clean and preprocess text data, the mathematical intuition behind TF-IDF, and how to measure vector distances using cosine similarity. Furthermore, I learned the importance of model deployment and persistence—saving trained matrices and vectorizers to create a responsive, user-facing search tool rather than a script that has to recompute everything upon every execution.

6. Screenshots

```
F:\Downloads\AIML>C:/Users/skaro/AppData/Local/Python/pythoncore-3.14-64/python.exe f:/Downloads/AIML/IDF-matrix.py
Libraries imported.
Fetching/Loading 20 Newsgroups data...
Dataset loaded and shuffled consistently.
Number of documents: 18846
Number of classes: 20

Creating TF-IDF Vectorizer and Matrix...
TF-IDF matrix created.
Shape: (18846, 51840)
Matrix shape matches target length.

Performing immediate classification sanity check (expecting >= 75.0% accuracy)...

Sanity Check Accuracy: 0.7692
Sanity Check PASSED: TF-IDF matrix appears reasonable for classification.

Classification Report (Sanity Check):
                          precision    recall  f1-score   support

             alt.atheism       0.73      0.56      0.63       160
           comp.graphics       0.73      0.74      0.74       195
 comp.os.ms-windows.misc       0.76      0.70      0.73       197
comp.sys.ibm.pc.hardware       0.64      0.68      0.66       196
   comp.sys.mac.hardware       0.83      0.79      0.81       193
          comp.windows.x       0.87      0.85      0.86       198
            misc.forsale       0.81      0.78      0.80       195
               rec.autos       0.82      0.77      0.79       198
         rec.motorcycles       0.89      0.85      0.87       199
      rec.sport.baseball       0.92      0.83      0.87       199
        rec.sport.hockey       0.58      0.92      0.71       200
               sci.crypt       0.81      0.90      0.85       198
         sci.electronics       0.84      0.75      0.79       197
                 sci.med       0.93      0.87      0.90       198
               sci.space       0.84      0.80      0.82       197
  soc.religion.christian       0.59      0.88      0.71       199
      talk.politics.misc       0.73      0.54      0.62       155
      talk.religion.misc       0.84      0.25      0.39       126

                accuracy                           0.77      3770
               macro avg       0.78      0.76      0.76      3770
            weighted avg       0.78      0.77      0.77      3770


Saving verified TF-IDF components and aligned targets...
Components saved successfully:
  - tfidf_vectorizer_verified.joblib
  - tfidf_matrix_verified.joblib
  - target_aligned.joblib
  - target_names_aligned.joblib
```

```
F:\Downloads\AIML>C:/Users/skaro/AppData/Local/Python/pythoncore-3.14-64/python.exe f:/Downloads
/AIML/IDF-search.py
Libraries imported.
Loading verified components for search and evaluation...
Loaded vectorizer, matrices, targets, and names successfully.

Reloading and shuffling texts from cache 'dataset' for context...
Texts reloaded and aligned successfully.

Components ready for search and evaluation.

--- Interactive Search ---
Type your query or 'quit' to exit.
Enter search query: war

--- Top 5 Results for 'war' ---

1. Index: 10667, Score: 0.4785, Category: talk.religion.misc
   Snippet: UN Resolution 666 guarantees humanitarian aid will get into Irag during
the Gulf War.  Is this war over? Is aid getting in, or are they still
trying to smoke out Saddam?  Is this the Middle East?  Are we talking
religious war here?  Am I ranting?
```

```
2. Index: 8176, Score: 0.3934, Category: talk.politics.mideast
   Snippet:


I would hope that if you intend to have a reasonable discussion you might
wait until I express an opinion before deciding I should be flamed for it.
As for 'war' I am not sure how I would define it. If you just look at attacks
on villages then there is no way of deciding when it started. Would you
count the riots in the 20's and 30's? Violence but not war. I personally
think that 'war', as opposed to civil disturbance or whatever, requires
organisation, planning and some measure of regualr or...

3. Index: 16316, Score: 0.3857, Category: soc.religion.christian
   Snippet: Hi there netters,
I have a question I would very much like to see some discussion on:
Is there such a thing as a 'justifible' war?

What I would love to see it some basis from scripture for either: "All war
is wrong", or "Some war is justifiable".

To get things started I would like to outline why I am asking the
question. In my high school days I had been quite involved in the the New
Zealand Cadet Forces (This is a bit like ROTC from what I understand of
it, but with a lot more empha...
```

```
4. Index: 7544, Score: 0.3790, Category: talk.politics.mideast
   Snippet:

Not to muslims :)



That could be arguable.. (for bandwith and flames sake, I wont say more)



It is a civil war, but the serbian generals who allow rape are not fighting
It is a civil war, but the serbian generals who allow rape are not fighting
It is a civil war, but the serbian generals who allow rape are not fighting
It is a civil war, but the serbian generals who allow rape are not fighting         oth..
It is a civil war, but the serbian generals who allow rape are not fighting
fair.  Yah I know .. war is hell  ..  those serbian generals are propretuating  It is a civil war, but the s
an generals who allow rape are not fighting      oth..
fair.  Yah I know .. war is hell  ..  those serbian generals are propretuating  It is a civil war, but the s
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generals who allow rape are not fighting    th..
fair.  Yah I know .. war is hell  ..  those serbian generals are propretuating nes and bordres .. money .. p
 .. fear ..
It is a civil war, but the serbian generals who allow rape are not fighting
fair.  Yah I know .. war is hell  ..  those serbian generals are propretuating
It is a civil war, but the serbian generals who allow rape are not fighting
fair.  Yah I know .. war is hell  ..  those serbian generals are propretuating
It is a civil war, but the serbian generals who allow rape are not fighting
It is a civil war, but the serbian generals who allow rape are not fighting
fair.  Yah I know .. war is hell  ..  those serbian generals are propretuating
both..

Lines and bordres .. money .. power .. fear ..

Let's change canadian bordres while we are at it :)
I see this as civvil war..
(sp borders)



Possibly..  I do agree that it is a civil war, which makes the donation of
```

```
5. Index: 18420, Score: 0.3532, Category: talk.politics.misc
   Snippet:
Go for it.  I have yet to see anybody justify the
prohibition on drugs and the ensuing War On Drugs.  In the world of
*.politics here on Usenet, it is YOU that is crazy.  ANYBODY--who gives
the matter any thought beyond reading headlines---cannot justify this
atrocity, this all out war on individual rights.

Just _TRY_ to justify the War On Drugs, I _DARE_ you!


Once again, in chorus: WHY is this "stupid"?


The above paragraph is gibberish--that all I can make of it...


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Enter search query:
```